

Prathyush Kumar Reddy Lebaku | Data Scientist

Location: Texas, USA | Mail: Prathyush.lebaku5@gmail.com | Ph.:+1 832-856-6716 | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

PROFESSIONAL SUMMARY:

Data Scientist with 3+ years of experience leveraging machine learning, deep learning, and large language models (LLMs), NLP in healthcare, transportation and financial domains. Skilled in predictive/forecasting modeling, RAG systems with LoRA-adapted BioBERT, and big data tools (Spark, Hadoop, Azure, AWS). Proven impact includes a 70% improvement in clinical Q/A accuracy and a 30% reduction in loan defaults with expertise in Power BI, A/B testing, and SHAP explainability to deliver actionable insights, regulatory compliance, and enterprise-scale adoption.

EDUCATION:

Master of Science in Engineering Data Science - University of Houston, Main Campus Texas, USA – 3.91/4.0

Bachelor of Science in CSE with Specialization in Data Science - Vellore Institute of Technology, Vellore, Tamil Nadu, India

PROFESSIONAL EXPERIENCE:

McKesson – TX | Data Scientist | August 2024 – Present

Project Name: IntelliCare Clinical AI Suite

- Improved clinical Q/A chatbot by applying LoRA on BioBERT in RAG pipeline (Azure AI Foundry, LangChain, FAISS) over 300K+ documents, achieving 95% relevance (nDCG + recall@k), cutting latency by 50%, and boosting GPT-4o answer quality 70%.
- Enhanced brain tumor diagnostics with TensorFlow pretrained CNNs (transfer + ensemble learning) and heavy augmentation on 100K+ MRI/CT scans, raising accuracy 25%, lowering false positives 30%, and delivering ROC-AUC 0.87.
- Improved early detection of chronic diseases by 35% using risk prediction models (Random Forest, XGBoost, Artificial Neural Networks) on 2M+ patient records, achieving 92% Precision and ROC-AUC 0.83 by feature engineering and balancing.
- Reduced large scale healthcare data processing time by 40% by engineering Talend-based ETL pipelines on Azure to securely ingest and transform 10TB+ of EHR and claims data, maintaining HIPAA-compliance and accelerating downstream AI/ML workflows.
- Enabled 5,000+ clinicians with reliable AI services by productionizing models as containerized APIs on Azure AKS/ACI with CI/CD and autoscaling, achieving 99.99% uptime and reducing run costs by 30%.

HCLTech – India | Data Scientist | April 2021 – July 2023

Project Name: Intelligent Credit Risk Assessment and Fraud Prevention System

- Reduced loan defaults 30% by engineering credit risk prediction models with XGBoost and Random Forest (Scikit-learn) on 5M+ borrower records, achieving 91% precision and ROC-AUC 0.85, enabling financial institutions to proactively flag high-risk borrowers.
- Optimized liquidity management and reserve planning 25% by developing ARIMA and LSTM (PyTorch) models to forecast loan defaults, transaction volumes, and cash flow trends, achieving MAPE of 8% and improving forecasting accuracy by 20% over baseline.
- Improved credit card fraud detection recall to 93% by applying PyTorch-based LSTM Autoencoders on sequential transaction data, reducing false negatives 32% and F1-score of 89%.
- Partnered with ML engineering team to deploy end-to-end ML pipelines leveraging Azure services (Azure ML, Azure Data Factory, Azure Databricks, AKS) ensuring model reproducibility, automated retraining, and scalable analytics solutions for enterprise-wide adoption.
- Collaborated with a data engineering team to cut data preparation time 60% by designing and automating scalable ETL pipelines using Apache Airflow and Spark on Hadoop, enabling ingestion and preprocessing of millions of financial and loan transactions.
- Enhanced adoption of fraud alerts and credit risk insights by conducting A/B testing and delivering Tableau/Power BI dashboards, while applying causal inference and SHAP interpretability to explain model decisions to regulators, improving decision-making speed 30% and ensuring Basel II/III and RBI compliance.

TECHNICAL SKILLS:

Programming & ML: Python (Pandas, NumPy, scikit-learn, PyTorch, TensorFlow), XGBoost, Random Forest, ARIMA, LSTM, Autoencoders, Statistical Modelling, SHAP, SQL (T-SQL), PySpark, R, Geospatial data

Generative AI & NLP: Transformers (Hugging Face), NLTK, Spacy, RAG (LangChain, FAISS, AutoGen), BioBERT, n8n, OpenCV

Data Engineering: ETL Pipelines (Talend, Apache Airflow, Spark, Hadoop), Azure Data Factory, Data Preprocessing & Integration

Cloud & Deployment: Azure Cloud (Azure ML, AKS, ADF, Blob Storage), Containerization (Docker/Kubernetes), AWS (Sagemaker, S3), dbt, Git/GitHub, CI/CD, Agile/Scrum, Snowflake

Data Visualization: Power BI (DAX), Interactive Dashboards, A/B Testing Analytics, Tableau, Matplotlib, Seaborn

Healthcare & Finance Domain: Healthcare (EHR, DICOM, FHIR/HL7, ICD-10, CPT, HIPAA); Finance (risk/fraud); Basel II/III, RBI

CERTIFICATIONS:

- Microsoft Power BI Associate (PL-300)
- Microsoft Azure Data Scientist Associate (DP-100)
- Deep Learning (by Andrew Ng)
- AWS Cloud Practitioner (Amazon)

PROJECTS and PUBLICATIONS

Deep Learning for Pavement Condition Evaluation Using Satellite Imagery.

- Addressed the challenge of costly and labour-intensive pavement inspections by developing a deep learning framework that evaluates road conditions using satellite imagery (NTTA, Usa). Used AWS SageMaker with transfer learning, ensembling, and heavy augmentation on 30k+ images, achieving 93% accuracy, F1 0.93, AUROC 0.89; deployed as a SageMaker endpoint, with artifacts stored in S3.

Cybersecurity-Focused Anomaly Detection in Connected Autonomous Vehicles Using ML.

- Engineered a cybersecurity framework using Random Forest and Stacked LSTM models to identify anomalies in Connected Autonomous Vehicle (CAV) networks. Validated the models on a simulated dataset generated using matplotlib via the Intelligent Driver Model (IDM), successfully detecting attack patterns with R^2 scores of 0.9830 and 0.9998, respectively.

Assessing the Influence of Pavement Performance on Road Safety Through Crash Frequency and Severity Analysis.

- Developed a geospatial data pipeline using BeautifulSoup, Shapely, and GeoPandas to integrate crash coordinates with pavement condition dataset from the IOWA DOT, performing spatial joins and buffer analysis to study roadway impacts; applied ML, statistical inferencing and Statistical modelling (pseudo R^2 : 0.73) to evaluate how pavement roughness, and friction influence crash frequency and severity.